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ORIGINAL ARTICLE

Sensitivity and specificity of presumptive tests for blood, saliva and semen

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Abstract
Purpose Despite their wide use, the limits of presumptive tests can be poorly understood. The aim of this study was to investigate the specificity and sensitivity of conventional, as well as innovative, presumptive tests for blood, semen and saliva.
Methods We investigated Kastle–Meyer (KM) and leucomalachite green (LMG) tests for blood with regard to their sensitivity and specificity in the presence of oxidizing (hypochlorite) and anti-oxidizing (ascorbic acid) agents. The suitability and specificity of the red starch paper (RSP) test for saliva was assessed. Finally, the inhibitory effect of detergent on the acid phosphatase (AP) test for semen was investigated along with possible cross reactions to tea stains.
Results Our results confirm previous findings of higher sensitivity and specificity of the KM test compared to LMG test for blood. Contrary to previous studies, no statistically significant difference was observed in the sensitivity of the tests between dry and wet stains. The novel RSP test was found to successfully detect saliva. We demonstrated that acid phosphatase (AP) testing for semen is possible on used RSP. A common multipurpose detergent had an inhibitory effect on AP tests. False positive results were obtained from tea stains. Testing different sorts of tea (black, green and herbal teas) revealed that only *Camellia* varieties produce positive result with the AP test, due to AP being present in the plants.
Conclusions From our results we conclude that specific knowledge of each test, including substances that may affect the test outcome, is imperative to ensure correct interpretation of presumptive test results.

Keywords Biological presumptive test · Sensitivity · Specificity · Acid phosphatase · Tea leaves

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Introduction

The identification of the type of body fluid present in a biological stain has important implications on the processing and preparation of the sample for subsequent forensic testing. Furthermore, in some circumstances the presence or absence of certain body fluids can directly aid police investigations, for example in cases of alleged sexual assault.

Conventional presumptive tests are often based on either chemical reactions, such as the Kastle–Meyer test for the presence of blood [1–3], or enzymatic reactions, such as the acid phosphatase test for the presence of seminal fluid [4], leading to a color change. Such testing strategies provide easy-to-use, low-cost and time efficient tools for forensic science. However, detailed knowledge of specificity and sensitivity of presumptive tests is crucial for

correct interpretation and inclusion of the test results into forensic investigations.

Sensitivity of presumptive tests are often reported with vastly different values for the same reagents, especially in the case of presumptive blood tests [5], which has been theorized to occur based on the testing conditions, i.e. wet stains versus dry stains [6]. The interpretation of results can also be challenging because cross reactivity with common household chemicals or natural products is known for many tests [5, 7] and false negative test results due to inhibition can obscure the true identity of a body fluid. For example, the presence of acid phosphatase in the tea leaf was demonstrated by Tirimanna [8], causing false positive results with the acid phosphatase test, but in 1994 further testing did not find any false positive reactions of the acid phosphatase test with tea [7].

Thus, the aim of this work was to enhance knowledge and experience in the interpretation of commonly used, as well as innovative, presumptive tests. Commonly used tests comprise the Kastle–Meyer-test and leucomalachite-green-test for the presence of blood and the acid phosphatase test for the presence of semen. The innovative test included in this work is the red starch paper test for the presence of saliva as recently described by Martin et al. [9]. Data were collected in 57 replicates under identical conditions with identical chemicals and substrates. All data were obtained during the postgraduate course in Forensic Science at the University of Strathclyde. A total of 57 students took part in this study. This provided the unparalleled possibility of performing each test in 57 replicates under identical laboratory conditions and with identical substrates and reagents.

Materials and methods

All experimentation was carried out in the Forensic Science Postgraduate Laboratories at the University of Strathclyde. Benches were cleaned using a Trigene solution and gloves were changed prior to, and in-between, the different stages of experimentation.

Substrates

Bovine blood was obtained from a local abattoir. Both blood presumptive tests included in this study are based on oxidization reactions. To test for false positive results and inhibition of these tests, the effects of a strong oxidant (5 % hypochlorite) and a strong anti-oxidant (0.1 M ascorbic acid) were analyzed. Furthermore, a ferrous salt (10 % ferrous sulfate) was also tested in the absence of blood because it is known to act as a strong oxidant and may produce a color reaction even before the addition of hydrogen peroxide.

Saliva and semen samples were collected from healthy donors with informed consent. Samples from different donors were pooled to provide a large amount of a uniform sample for all test replicates. Pooled samples were spotted onto filter paper or used in dilution series.

To investigate positive reactions of the acid phosphatase test with tea, the following tea samples were prepared: *Camellia sinensis (C. s.) sinensis* ("Darjeeling") and *C. s. assamica* ("Assam") teas were obtained as fermented ("black") and unfermented ("green") teas, respectively. Furthermore, flavored black tea ("Indian Chai"), Rooibos and herbal teas (chamomile, fennel and peppermint) were included in the study. Teas were brewed in 200 mL of boiling water according to the manufacturers' instructions and a swab was submerged into the liquid for testing. In addition to *C. sinensis* varieties, leafs from the closely related common garden plants *C. japonica* and *Erica spec.* were obtained, covered in 300 μL water and vortexed for 2 min before testing 15 μL of the solution.

A semen positive control (dilution 1:1,000) was obtained from a healthy male volunteer.

Kastle–Meyer-test and leucomalachite-green-test

Reagents for the Kastle–Meyer (KM)-test, also known as phenolphthalein test, and the leucomalachite-green (LMG)-test, were prepared as previously described [5]. A tenfold series of blood dilutions were prepared ranging from neat to 10^{-7} using whole blood and distilled water and were thoroughly mixed to ensure a uniform distribution of the blood components.

To test the dilutions, 20 μL of the sample was pipetted on to a piece of filter paper. This was then tested by adding one drop of KM or LMG reagent followed by one drop of hydrogen peroxide (3 %). A positive reaction was indicated by a color change (either pink or green for KM and LMG, respectively) at the site of the stain. This was repeated for each of the dilutions. Additionally, to investigate the effect of wet and dry stains, 20 μL of the sample was pipetted on to a sterile tile and allowed to dry before being tested in the manor previously described.

To test for the influence of oxidizing and anti-oxidizing agents, one drop of hypochlorite or ascorbic acid were added to the filter paper prior to the deposition of the blood dilution sample. The samples were then tested in the manner previously described. Furthermore, the reactivity of both methods with ferrous sulfate in the absence of blood was tested.

The χ^2 test in a two-by-two contingency table was used to assess differences between the different tests and sets of testing conditions.

Red-starch-paper-test

A 0.3 % red starch (Megazyme International Ireland Ltd, Dublin, Ireland) solution was prepared and applied to filter paper as described by Martin et al. [9]. To test for the presence of α-amylase from saliva both the red starch paper and the saliva stain were moistened with distilled water. The stain was pressed onto the moistened red starch paper, placed in between two plates of glass, wrapped in moistened tissues and incubated between 30 and 90 min at 20 °C. The samples were then removed from the incubator and observed. A positive reaction was indicated by a pale area appearing on the red starch paper corresponding to the stain.

Samples of blood and semen were also tested using red starch paper according to the previously described protocol. To test whether the acid phosphatase procedure can be used to test for the presence of semen after testing for the presence of saliva [9], the red starch paper was sprayed with acid phosphatase as described below.

Acid-phosphatase-test

The acid phosphatase (AP) reagent contained two stock solutions: stock solution A was prepared from 1 g of orthodianisidine tetrazotised, 20 g sodium acetate, 10 mL acetic acid, and 100 mL of water. Stock solution B was prepared from 0.8 g sodium alpha-naphthyl phosphate and 10 mL of water. The AP reagent was then prepared by mixing 10 mL of Stock Solution A, 89 mL of water, and 1 mL of Stock Solution B. The test was performed by pressing a moistened filter paper onto the stain for 1 min and spraying with AP reagent. AP was directly sprayed on the red starch paper to test for seminal fluid after saliva testing to obtain a combined semen/saliva test. A positive result was indicated by a purple color appearing within 2 min corresponding to the site of the stain. To test for inhibition, a fresh semen stain was sprayed with 1 % Teepol multi-purpose detergent (Teepol, Orpington, UK) before AP testing.

To test for a false positive reaction with different tea varieties, the Phosphatesmo KM test strips (Macherey–Nagel, Düren, Germany) were used by pressing the moist swabs onto the test strip. A positive result was recorded when a purple color appeared within 2 min.

Results

Blood presumptive tests

All dilution series results can be found in Table 1. The LMG test gave 100 % positive results in all experiments down to a dilution of 10^{-2}; and the KM test produced 100 % positive results down to 10^{-3}. At the final dilution factor of 10^{-7}, KM still showed 54.4 % positive results while LMG showed 33.3 % positive results ($p < 0.05$).

Table 1 Results of the Kastle–Meyer (KM) and the leucomalachite-green (LMG) tests obtained with dilutions of bovine blood: pure, in the presence of ascorbic acid, and in the presence of hypoclorite, respectively

	Blood		Blood with ascorbic acid		Blood with hypochlorite	
	KM (%)	LMG (%)	KM (%)	LMG (%)	KM (%)	LMG (%)
Neat	100	100	54.39	21.05	96.49	92.98
10^{-1}	100	100	42.11	10.53	96.49	92.98
10^{-2}	100	100	26.32	10.53	94.74	91.23
10^{-3}	100	91.23	12.28	1.75	91.23	77.19
10^{-4}	94.74	73.68	10.53	1.75	73.68	64.91
10^{-5}	87.72	50.88	8.77	1.75	61.40	57.89
10^{-6}	73.68	36.84	5.26	1.75	57.89	56.14
10^{-7}	54.39	33.33	5.26	1.75	54.39	52.63

Numbers given represent the percentage of positive results observed in 57 tests

Chi squared analysis showed that the KM and LMG tests were significantly different when tested over the dilution series for both wet and dry stains. Comparison of wet versus dry stains showed that there was no difference within the dilution series of the same test ($p < 0.005$ for KM and $p < 0.001$* for LMG).

In the presence of ascorbic acid, both blood presumptive tests showed signs of inhibition with KM showing 54.4 % and LMG 21.1 % ($p < 0.05$) positive results in the neat blood sample compared to 100 % positive results without ascorbic acid.

Addition of hypochlorite to the dilutions had an inhibitory effect on the KM test when compared to the untreated dilutions. When the hypochlorite treated samples were compared between the two tests, there was no longer any significant difference between the observed sensitivities ($p = 0.055$). The LMG test showed higher percentages of positive results in the higher dilutions from 10^{-5} to 10^{-7} compared with untreated blood.

The 10 % ferrous sulfate solution used in the absence of blood resulted in the KM test showing 85.3 % false positive results while with the LMG test, 73.5 % false positive results were observed.

Saliva and combined semen/saliva test

The red starch paper prepared in-house showed 91–97 % positive results (Table 2). Different incubation times were tested but no statistically significant differences were observed between time points. False positive results were observed for blood (14–25 %) and semen (7–15 %) (Table 2).

Table 2 Results of the red starch paper (RSP) test and the subsequently performed acid phosphatase test (AP) on red starch paper

Incubation	RSP (%)	AP on RSP (%)
Saliva (min)		
30	94.64	5.45
60	96.43	7.27
90	91.07	5.45
Semen (min)		
30	14.29	87.27
60	25	83.64
90	25	80
Blood (min)		
30	8.93	3.64
60	14.29	3.64
90	7.14	1.82

Numbers given represent the percentage of positive results observed in 57 tests

The acid phosphatase test on red starch paper showed false positive results with both blood and saliva (Table 2). When tested with semen, between 80 and 88 % positive results were observed, with the highest accuracy occurring with the shortest incubation time (Table 2).

Semen presumptive test

Semen stains that were tested with AP reagent gave 100 % positive results. After spraying the stain with detergent, only 16.67 % positive results were obtained.

To test for false positive results, tea stains were tested with the AP reagent and a total of 75 % positive results were obtained. Tea made from *C. s. sinensis* ("Darjeeling") and *C. s. assamica* ("Assam") both fermented (black tea) and unfermented (green tea) showed a strong purple color change (Fig. 1). Chai tea showed a strong purple and orange color change.

Camomile and fennel tea gave a negative result showing no color change. Rooibos tea showed an orange discoloration of the test strip similar to the color of the tea itself.

Due to the false positive reactions of the acid phosphatase test only with tea made from *Camellia sinensis* varieties we tested closely related garden plants (*C. japonica* and *Erica spec.*). Both displayed the same false positive color changes (Fig. 1).

Discussion

Blood presumptive tests

The KM test seems to be slightly more sensitive than the LMG test, which becomes particularly pronounced in the higher dilutions (Table 1). This was not found in a previous study [5], but in that study the dilution series started at $1{:}10^{-5}$ and most samples failed to react past the first dilution.

Testing of wet or dry stains did not reveal any differences in sensitivity within the same test, the first time this has been empirically tested. This demonstrates that in expected case situations the analysis of a wet or dry stain should have no impact on the sensitivity of either KM or LMG. Grodskey et al. [6] theorized that testing of a wet or dry stain could significantly impact the observed sensitivity of a test as dilution of the reagents would occur when tested with a wet sample. This may hold true if testing in a large volume of fluid (such as a cuvette) but in practical terms, as demonstrated in this study, will not have an impact on the results obtained.

Addition of ascorbic acid caused both the KM and LMG tests to fail and can be seen as a strong inhibitor of both KM and LMG with the LMG test being more affected. The inhibition is due to the strong anti-oxidizing characteristic of ascorbic acid. This has implications for case work scenarios where biological stains are found in close proximity to ascorbic acid sources, such as vegetables, fruits or juices.

Hypochlorite is a strong oxidizing agent and as such is expected to produce false positive reactions with both blood tests. The trend towards more positive reactions of LMG with highly diluted blood samples in the presence of hypochlorite could be interpreted as a result of the reaction of bleach with LMG, oxidizing it to form the colored component. As hypochlorite is a component of many bathroom and kitchen cleaning agents as well as some washing powders, the effects of hypochlorite has to be kept in mind when interpreting presumptive test results from, for example, swabs taken from bathroom surfaces or stains on white fabrics, because these items may have been treated with hypochlorite.

The high percentage of false positive results with ferrous sulfate is surprising because ferrous sulfate is known to exhibit a very different color (orange–brown) from what is expected to occur with the tests. Furthermore, the orange–brown color appears before hydrogen peroxide is added. This has been previously observed with ferrous, as well as other metal, salts and therefore there should be no misinterpretation of results [5]. We noticed, however, that in addition to the color change known from ferrous sulfate the specific pink and turquoise color for KM and LMG, respectively, appears immediately after addition of hydrogen peroxide and forms a ring, or halo, around the orange centre (Fig. 2). This additional color change is obviously a false positive reaction that can lead to misinterpretation of the test in a forensic setting. Ferrous salts can be present in sinks, showers and baths due to old/eroding pipes.

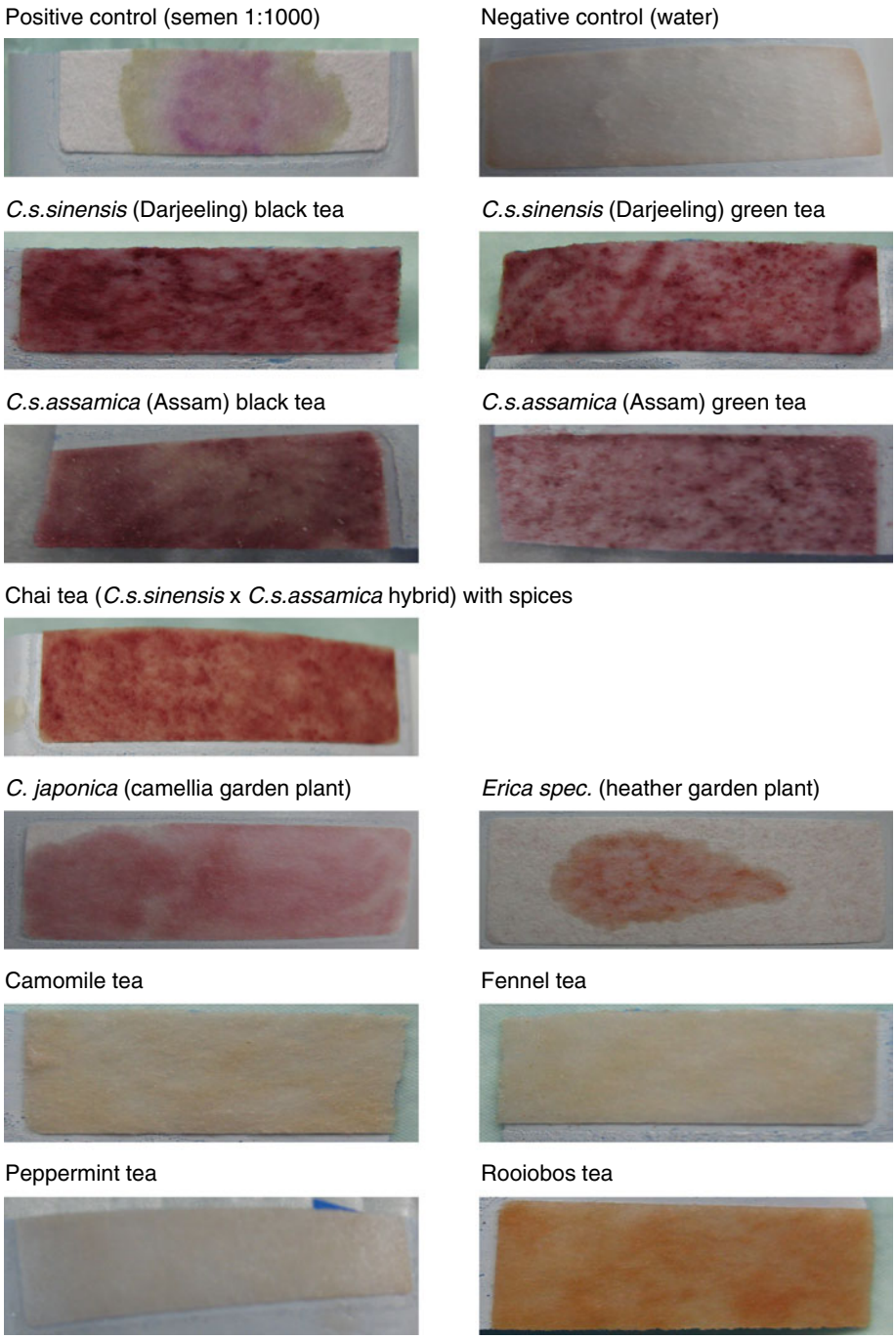


Fig. 1 Acid phosphatase test (using Phosphatesmo strips) with different tea varieties and two common garden plants. The typical color is visible in all Camellia varieties, but not in herbal teas

Saliva and combined semen/saliva test

In general, the red starch paper prepared in-house seems to be suitable for saliva presumptive testing. The number of false negative results is rather high considering undiluted saliva was used. This might be due to a difficulty in the interpretation of the test: a pale pink colored paper exhibiting a slightly paler area. Also, the relatively high number of false positive results for blood and particularly for semen shows that further extensive validation is necessary to make this test fit for use in forensic case work.

The acid phosphatase test on red starch paper generally worked well (Table 2), confirming findings by Martin et al. [9]. The high number of false positive and false negative results observed, as detailed in Table 2, show that this method is not yet fit for use in forensic case work and needs further validation before being able to replace well validated methods such as the Phadebas method [10].

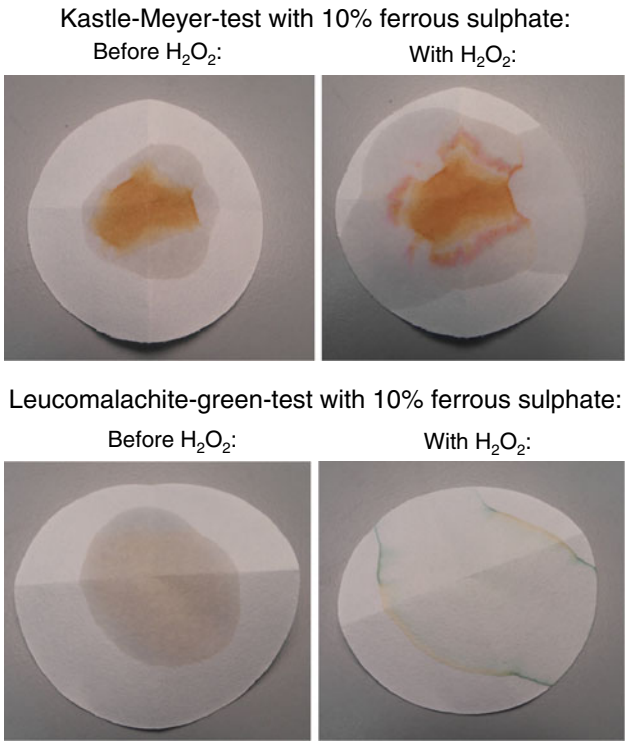


Fig. 2 Leucomalachite Green and Kastle–Meyer-Test with 10 % ferrous sulfate in absence of blood. The typical test colors are visible around the edges of the wet stain after addition of hydrogen peroxide

Semen presumptive test

Semen stains tested in the presence of detergent showed strong inhibition with only 16.67 % positive results observed. Teepol, the detergent used in this study, destroys cell walls and denatures proteins [11] and thus, the acid phosphatase seems to be damaged to an extent that reaction with α-naphthyl phosphate becomes impossible. Teepol, however, is a very strong detergent and results obtained here cannot directly be transferred to other detergents, such as washing powder. Sackiewicz et al. [12] found that even after washing semen stains with a conventional laundry detergent, the stains were still visible with alternate light sources. Therefore further testing investigating the effect of other detergents is needed to confidently exclude a sample showing negative results.

The Phosphatesmo test strip, a ready-made testing device relying on the well-established AP testing principle, was investigated. The Phosphatesmo strip allows acid phosphatase testing at a scene without the need for solutions or spray bottles. We examined semen as well as tea, which has previously been shown to produce false positives with AP. The test strips showed 100 % positive results when tested with semen. When tested with tea stains a total of 75 % false positive results were obtained. The presence of acid phosphatase in tea leaves has been previously described [8].

We further investigated which types of tea would produce a false positive result (Fig. 1). Both black and green Darjeeling and Assam teas showed a strong purple color change that, when compared to the positive control, was slightly darker. This is most likely due to the brown color of the tea itself adding its color to the positive purple of the test strip. Chai tea showed a strong purple and orange color change which can be interpreted as a combination of the acid phosphatase test color and the colors of the spices, such as cinnamon, added to the *C. sinensis* hybrids used for this tea.

The discoloration of the test strip by Rooibos tea can most likely be attributed to tea coloring the test strip rather than the test strip reacting with any of the components of the tea.

Thus, we conclude that false positive reactions of the acid phosphatase test occur with tea made from *Camellia sinensis* varieties only. Testing of closely related garden plants (*C. japonica* and *Erica spec.*) showed the expected color changes. In alleged sexual assault cases it is therefore possible to misinterpret a tea stain found on fabric for a semen stain. AP testing out doors or in the presence of certain plants may also produce a false positive result.

Conclusion

With this work we aimed to reinforce the knowledge on the limited sensitivity and specificity of commonly used as well as innovative methods for forensic body fluid identification. We demonstrated that in casework situations the state (wet or dry) of a stain will not affect the presumptive blood tests KM or LMG. Test results can easily be misinterpreted due to the presence of very common household items such as cleaning agents, food or tea.

Even though innovative techniques such as RNA profiling [13] and DNA methylation analysis [14] can overcome some of the limitations of the conventional chemical tests the authors are convinced that such novel tests will not entirely replace conventional presumptive testing in the near future. This is due to the high costs of the novel techniques and a lack of commercially available kits, as well as guidelines concerning quality and interpretation issues.

Key Points

1. The Kastle–Meyer presumptive test for blood seems to be slightly more sensitive and less prone to false positive reactions caused by hypochlorite (a strong

oxidizing agent) than the alternative method, the leucomalachite green test.
2. In both blood presumptive tests, no differences in the sensitivity of wet versus dry stains were observed.
3. The red starch paper test for the presence of saliva did work well in our study and subsequent acid phosphatase testing on the test strip was possible. However, a relatively high number of false positive and false negative results suggest that further validation of this method is needed.
4. Acid phosphatase (AP) testing revealed a surprisingly high percentage of false positive results from tea stains. Further investigation revealed that AP reacted positively with green and black teas of the *Camellia sinensis* family as well as closely related garden plants. Other types of tea, such as herbal teas and Rooibos, showed no reaction with the AP test.

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References

1. Kastle JH, Shedd OM. Phenolphthalin as a reagent for the oxidizing ferments. Am Chem J. 1901;26:526.
2. Meyer E. Beiträge zur Leukocytenfrage. Münchner Medizinische Wochenschrift. 1903;50:1489–93.
3. Cox M. A study of the sensitivity and specificity of four presumptive tests for blood. J Forensic Sci. 1991;36:1503–11.
4. Kind SS. The use of the acid phosphatase test in searching for seminal stains. J Crim Law Criminol Police Sci. 1957;47:597–600.
5. Tobe SS, Watson N, Daéid NN. Evaluation of six presumptive tests for blood, their specificity, sensitivity, and effect on high molecular-weight DNA. J Forensic Sci. 2007;52:102–9.
6. Grodsky M, Wright K, Kirk PL. Simplified preliminary blood testing-an improved technique and a comparative study of methods. J Crim Law Criminol Police Sci. 1951;42:95–104.
7. Hooft PJ, van de Voorde HP. Interference of body products, food and products from daily life with the modified zinc test and the acid phosphatase test. Forensic Sci Int. 1994;66:187–96.
8. Tirimanna AS. Acid phosphatases in the tea leaf. Experientia. 1972;28:633.
9. Martin NC, Clayson NJ, Scrimger DG. The sensitivity and specificity of red-starch paper for the detection of saliva. Sci Justice. 2006;46:97–105.
10. Hedman J, Dalin E, Rasmusson B, Ansell R. Evaluation of amylase testing as a tool for saliva screening of crime scene trace swabs. Forensic Sci Int Genet. 2011;5:194–8.
11. Bright JA, Cockerton S, Harbison S, Russell A, Samson O, Stevenson K. The effect of cleaning agents on the ability to obtain DNA profiles using the Identifiler™ and PowerPlex® Y multiplex kits. J Forensic Sci. 2011;56:181–5.
12. Sackiewicz A, Niemcunowicz-Janica A, Pepiński W, Skawrońska M, Szeremeta M, Ptaszyńska-Sarosiek I, Okłota M. Evaluation of visualization of biological stains with the use of alternative light source (ALS) for the purpose of genetic identification. Part II. Semen samples analysis. Arch Med Sadowej Kryminol. 2010;60:258–62.
13. Haas C, Hanson E, Ballantyne J. Capillary electrophoresis of a multiplex reverse transcription-polymerase chain reaction to target messenger RNA markers for body fluid identification. Methods Mol Biol. 2012;830:169–83.
14. Gršković B, Zrnec D, Vicković S, Popović M, Mršić G. DNA methylation: the future of crime scene investigation? Mol Biol Rep. 2013;40:4349–460.